Introduction to Statistical Machine Learning

ML & Deep Learning Course

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Outline

1. What is Machine Learning?
2. ML vs. Econometrics
3. Risk, Loss Functions, and ERM
4. Model Complexity and Generalization
5. Concentration Inequalities
6. Rademacher Complexity
7. VC Dimension
8. Overfitting and Sample Splitting

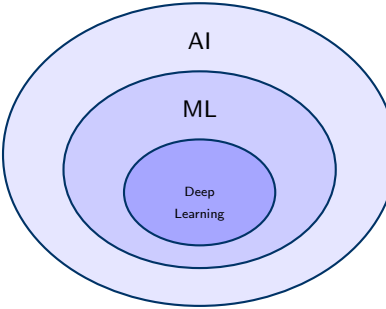
Artificial Intelligence and Machine Learning

Broad view

Artificial Intelligence: building systems that perform tasks normally requiring human intelligence.

Machine Learning: a sub-field of AI that gives computers the ability to *learn from data* without being explicitly programmed.

Deep Learning: a subset of machine learning based on artificial neural networks with multiple layers that mimic the human brain to analyze data.



Applications of Machine Learning

Classic domains (Classification and Regression):

- Spam / e-mail filtering (SVM)
- Handwriting & speech recognition (Gaussian Mixtures and Markov Models)
- Credit scoring, fraud detection (Random Forest, Boosting)
- Medical image diagnosis (Clustering)
- Recommendation systems

Modern, large-scale domains:

- Large Language Models (Transformers)
- Self-driving vehicles (NN Classification)
- Image Generation (Diffusion Models)
- Real-time ad targeting (Google, Meta)
- Drug and Protein Design (GFlowNets)

Why now?

1. **Data**: internet-scale datasets (billions of examples)
2. **Compute**: GPUs / TPUs make training feasible
3. **Algorithms**: better architectures and optimization

Timeline of AI, ML, and Deep Learning

1950s – 1970s
Early AI

Focus on symbolic logic and early perceptrons. Optimism eventually hit theoretical limits.

1970s & 1980s
The AI Winters

Periods of skepticism and funding cuts due to computational limits and inflated expectations.

1990s – 2000s
Statistical ML

Shift toward probability and optimization (e.g., SVMs, Random Forests). Rigorous, theory-driven models.

2010s – Present
DL Renaissance

Paradigm shift driven by **internet-scale data** and **GPUs**. Shift from manual feature engineering to automated, hierarchical feature extraction.

Methodological and Philosophical Differences

	Econometrics	**Statistical / ML**
Primary goal	Inference about parameters	Prediction / decision
Model role	Data Generating Process (DGP)	Prediction machine
Model selection	Theory-driven	Data-driven
Evaluation	p-values, confidence intervals	Out-of-sample loss
Interpretability	Central	Secondary (often)
Sample size	Often small ($n \ll p$ rare)	Can handle $n \gg p$ or $n \ll p$
Causal claims	Core mandate	Separate track (Causal ML)
Overfitting	Addressed via parsimony	Addressed via regularisation

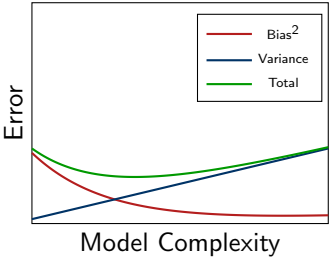
Breiman's "Two Cultures" (2001)

Statisticians use *data models*: assume stochastic data model, fit parameters. ML uses *algorithmic models*: treat nature as a black box, optimize predictive accuracy.

The Google Way of Doing Things

Scale changes the rules

- Norvig & Halevy (2009): "*The Unreasonable Effectiveness of Data*": simple models with massive data beat sophisticated models with small data.
- With $n \to \infty$, variance $\to 0$: bias becomes the binding constraint $\Rightarrow$ **expressive models** win.
- Engineering effort shifts from model design to **data pipelines**, **distributed training**, and **hyperparameter search**.



The Statistical Learning Setup

- **Input space**: $\mathcal{X}$ (feature vectors)
- **Output space**: $\mathcal{Y}$ (labels / targets)
- **Unknown distribution**: $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$
- **Training sample**: $S = ((x_1, y_1), \ldots, (x_m, y_m)) \overset{iid}{\sim} \mathcal{D}^m$
- **Hypothesis class**: $\mathcal{H}$ — the set of candidate predictors $h : \mathcal{X} \to \mathcal{Y}$
- **Loss function**: $\ell : \mathcal{H} \times (\mathcal{X} \times \mathcal{Y}) \to \mathbb{R}_{\geq 0}$

Fundamental question

How do we find $h \in \mathcal{H}$ that *generalises well* to unseen data from $\mathcal{D}$, given only S?

Classification vs. Regression

Classification

- Output space $\mathcal{Y}$ is **discrete**: $\{0, 1\}$.
- Goal: learn $h : \mathcal{X} \to \mathcal{Y}$
- Examples: spam detection, image classification, medical diagnosis
- Natural Target: Penalize binary mistakes (0-1)

Regression

- Output space $\mathcal{Y} \subseteq \mathbb{R}$ is **continuous**
- Goal: learn $h : \mathcal{X} \to \mathcal{Y}$
- Examples: house prices, demand forecasting, wage prediction
- Natural Target: Distance between the truth and prediction.

Unified view

Given sample $S = \{(x_i, y_i)\}_{i=1}^n \overset{iid}{\sim} \mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, find $h \in \mathcal{H}$ that minimizes some notion of error.

Loss Functions

0-1 Loss (Classification)

$$\ell_{\text{0-1}}(h,(x,y)) \;=\; \mathbf{1}[h(x) \neq y] \;=\; \begin{cases} 0 & \text{if } h(x) = y \\ 1 & \text{if } h(x) \neq y \end{cases}$$

Square Loss (Regression)

$$\ell_{\text{sq}}(h,(x,y)) \;=\; \big(h(x) - y\big)^2$$

Absolute Loss

$$\ell_{\text{abs}}(h,(x,y)) = |h(x) - y|$$

Logistic / Cross-Entropy Loss

$$\ell_{\log}(h,(x,y)) = \log\big(1 + e^{-y\,h(x)}\big)$$

The choice of loss encodes what errors we care about most.

Risk (True Error)

Definition (True Risk / Generalization Error)

The **risk** of a hypothesis $h \in \mathcal{H}$ with respect to distribution $\mathcal{D}$ and loss ℓ is:

$$L_{\mathcal{D}}(h) \;=\; \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\ell(h,(x,y))\big]$$

For the 0-1 loss: $L_{\mathcal{D}}(h) = \mathbb{P}_{(x,y)\sim\mathcal{D}}[h(x) \neq y]$.
Given training sample $S = ((x_1, y_1), \ldots, (x_m, y_m))$:

$$L_S(h) \;=\; \frac{1}{m}\sum_{i=1}^{m} \ell\big(h,(x_i,y_i)\big)$$

For 0-1 loss: $L_S(h) = \frac{1}{m}\big|\{i : h(x_i) \neq y_i\}\big|$.

The core problem

$\mathcal{D}$ is *unknown*. We observe only $L_S(h)$ and want to minimise $L_{\mathcal{D}}(h)$.

Bayes Optimal Predictor

Definition (Bayes Classifier)

The **Bayes optimal predictor** for distribution $\mathcal{D}$ under 0-1 loss is:

$$f_{\mathcal{D}}(x) \;=\; \mathbf{1}\big[\mathbb{P}[y = 1 \mid x] \geq \tfrac{1}{2}\big]$$

Theorem 1

No hypothesis $h : \mathcal{X} \to \{0, 1\}$ achieves lower true risk than $f_{\mathcal{D}}$:

$$L_{\mathcal{D}}(f_{\mathcal{D}}) \;\leq\; L_{\mathcal{D}}(h) \quad \forall h$$

For regression (square loss)

The Bayes optimal predictor is the conditional mean:

$$f_{\mathcal{D}}(x) = \mathbb{E}[y \mid x]$$

Achieves the **irreducible noise** $\mathbb{E}[\mathrm{Var}(y \mid x)]$ which no predictor can beat.

Empirical Risk Minimization (ERM)

ERM Learning Rule (Shalev-Shwartz & Ben-David, Def. 2.1)

Given hypothesis class $\mathcal{H}$ and sample S, the ERM learner outputs:

$$\mathrm{ERM}_{\mathcal{H}}(S) \in \operatorname*{argmin}_{h \in \mathcal{H}} L_S(h)$$

- **Inductive bias**: restricting to $\mathcal{H}$ encodes prior knowledge. (No Free-Lunch Theorem)
- **Realizability assumption**: $\exists\, h^\star \in \mathcal{H}$ s.t. $L_{\mathcal{D}}(h^\star) = 0$.
- **Agnostic setting**: no assumption on $\mathcal{H}$ relative to $\mathcal{D}$; we compete with $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$.

Danger: overfitting

Larger $\mathcal{H} \Rightarrow$ ERM fits training data better *but* may generalise poorly. **Smaller** $\mathcal{H} \Rightarrow$ may underfit.

Overfitting: The Core Pathology

Definition (Overfitting)

A learner **overfits** if it produces h with low training error but high true risk:

$$L_S(h) \approx 0 \quad \text{but} \quad L_{\mathcal{D}}(h) \gg 0$$

Canonical example

Let h_S be the predictor that memorises the training set:

$$h_S(x) = y_i \text{ if } x = x_i \in S, \quad 0 \text{ otherwise}$$

Then $L_S(h_S) = 0$, but for a uniform distribution, $L_{\mathcal{D}}(h_S) \approx 1/2$.

The Bias–Variance–Complexity Tradeoff

Excess Risk and Error Decomposition

The **excess risk** measures how much worse a hypothesis h performs compared to the best possible predictor in the universe (the Bayes optimal predictor $f_{\mathcal{D}}$):

$$\text{Excess Risk}(h) = L_{\mathcal{D}}(h) - L_{\mathcal{D}}(f_{\mathcal{D}})$$

For any $h \in \mathcal{H}$ returned by ERM, the excess risk decomposes into two components:

$$L_{\mathcal{D}}(h) - L_{\mathcal{D}}(f_{\mathcal{D}}) = \underbrace{L_{\mathcal{D}}(h) - \min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h')}_{\text{estimation error}} + \underbrace{\min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') - L_{\mathcal{D}}(f_{\mathcal{D}})}_{\text{approximation error}}$$

The tradeoff

- **Approximation error** (bias): how well $\mathcal{H}$ can express the true predictor. Decreases as $|\mathcal{H}|$ grows.
- **Estimation error** (variance): statistical fluctuation from finite S. Increases as $|\mathcal{H}|$ grows.

Richer $\mathcal{H}$ lowers approximation error but raises estimation error.

PAC Learning

Definition (Agnostic PAC Learnability) — Def. 3.3

$\mathcal{H}$ is **agnostic PAC learnable** (w.r.t. $\mathcal{Z}$, loss ℓ) if $\exists$ algorithm $\mathcal{A}$ and $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ s.t. for all $\varepsilon, \delta \in (0,1)$ and all $\mathcal{D}$ over $\mathcal{Z}$: running $\mathcal{A}$ on $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. examples returns h with probability $\geq 1 - \delta$:

$$L_{\mathcal{D}}(h) \;\leq\; \min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') + \varepsilon$$

Definition (Sample Complexity)

$m_{\mathcal{H}}(\varepsilon, \delta)$ is the **sample complexity**: minimum number of examples needed to guarantee ε-accuracy with confidence $1 - \delta$.

Sample Complexity for Finite Classes

Theorem 2 ((Finite Hypothesis Classes — Corollary 3.2))

Every **finite** *hypothesis class $\mathcal{H}$ is agnostic PAC learnable with sample complexity:*

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq \frac{\log(|\mathcal{H}|/\delta)}{2\varepsilon^2}$$

Proof sketch:

1. Apply Hoeffding's inequality to each $h \in \mathcal{H}$ separately.
2. Use union bound over all $|\mathcal{H}|$ hypotheses.
3. Conclude uniform convergence with the stated sample size.

Interpretation

Sample complexity grows *logarithmically* in $|\mathcal{H}|$ and *inverse-quadratically* in ε. Doubling accuracy requires $4\times$ more data.

Why Concentration Inequalities?

- We need to control $|L_{\mathcal{D}}(h) - L_S(h)|$: how well does the training error approximate the true risk?
- Tools: **Markov**, **Chebyshev**, **Hoeffding**, **McDiarmid**.

Lemma (Markov's Inequality)

For a non-negative random variable X and $a > 0$:

$$\mathbb{P}[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$$

Chebyshev's Inequality

For a random variable X with finite mean μ and variance σ^2:

$$\mathbb{P}[|X - \mu| \geq a] \leq \frac{\sigma^2}{a^2}$$

Both give *polynomial* decay in a. We want **exponential** concentration.

Hoeffding's Inequality

Theorem 3 ((Hoeffding's Inequality — Lemma 4.5))

Let $\sigma_1, \ldots, \sigma_m$ be independent random variables with $\mathbb{P}[a_i \leq \sigma_i \leq b_i] = 1$ and $\mathbb{E}[\sigma_i] = \mu_i$. Then for any $\gamma > 0$:

$$\mathbb{P}\left[\frac{1}{m}\sum_{i=1}^{m}\sigma_i - \frac{1}{m}\sum_{i=1}^{m}\mu_i \;\geq\; \gamma\right] \;\leq\; \exp\left(\frac{-2m^2\gamma^2}{\sum_{i=1}^{m}(b_i - a_i)^2}\right)$$

For i.i.d. $\sigma_i \in [a, b]$ with mean μ:

$$\mathbb{P}\left[\left|\frac{1}{m}\sum_{i=1}^{m}\sigma_i - \mu\right| \geq \gamma\right] \;\leq\; 2\exp\left(\frac{-2m\gamma^2}{(b-a)^2}\right)$$

Application

Set $\sigma_i = \ell(h, (x_i, y_i)) \in [0, 1]$. Then $\mathbb{E}[\sigma_i] = L_{\mathcal{D}}(h)$ and:

$$\mathbb{P}\big[|L_S(h) - L_{\mathcal{D}}(h)| \geq \varepsilon\big] \;\leq\; 2e^{-2m\varepsilon^2}$$

McDiarmid's Inequality

Theorem 4 ((Bounded Differences / McDiarmid))

Let $f(x_1, \ldots, x_m)$ be a function and suppose that changing a single argument changes f by at most c_i:

$$\sup_{x_1,\ldots,x_m,\, x_i'} \left| f(\ldots, x_i, \ldots) - f(\ldots, x_i', \ldots) \right| \;\leq\; c_i$$

Then for independent $x_1, \ldots, x_m$ and any $\varepsilon > 0$:

$$\mathbb{P}\big[f - \mathbb{E}[f] \geq \varepsilon\big] \;\leq\; \exp\left(\frac{-2\varepsilon^2}{\sum_{i=1}^m c_i^2}\right)$$

Key role

McDiarmid is the main tool for proving that the **Rademacher complexity** concentrates around its expectation, and for deriving uniform generalization bounds.

Uniform Convergence

Definition (Uniform Convergence Property)

$\mathcal{H}$ has the **uniform convergence property** with rate $m_{\mathrm{UC}}(\varepsilon, \delta)$ if: for every $\varepsilon, \delta \in (0, 1)$ and every $\mathcal{D}$, for $m \geq m_{\mathrm{UC}}(\varepsilon, \delta)$, with probability $\geq 1 - \delta$:

$$\sup_{h \in \mathcal{H}} \left| L_{\mathcal{D}}(h) - L_S(h) \right| \leq \varepsilon$$

Theorem 5 ((UC $\Rightarrow$ PAC Learnability))

If $\mathcal{H}$ has the uniform convergence property, then $\mathrm{ERM}_{\mathcal{H}}$ is an agnostic PAC learner with $m_{\mathcal{H}}(\varepsilon, \delta) \leq m_{\mathrm{UC}}(\varepsilon/2, \delta)$.

Proof: If $\sup_h |L_{\mathcal{D}}(h) - L_S(h)| \leq \varepsilon/2$, then:

$$L_{\mathcal{D}}(\hat{h}) \leq L_S(\hat{h}) + \tfrac{\varepsilon}{2} \leq L_S(h^\star) + \tfrac{\varepsilon}{2} \leq L_{\mathcal{D}}(h^\star) + \varepsilon$$

Rademacher Complexity: Motivation

- For **infinite** $\mathcal{H}$, union-bound arguments fail: $|\mathcal{H}| = \infty$.
- We need a complexity measure that captures the *effective* richness of $\mathcal{H}$.
- Idea: can $\mathcal{H}$ fit **pure noise**? Measure the correlation of $\mathcal{H}$ with random ± 1 labels.

Rademacher variables

$\sigma_1, \ldots, \sigma_m \overset{iid}{\sim} \mathrm{Uniform}(\{-1, +1\})$ are called **Rademacher** random variables.

Empirical and Population Rademacher Complexity

Definition (Empirical Rademacher Complexity — Def. 26.1)

Let $\mathcal{F} = \{z \mapsto \ell(h, z) : h \in \mathcal{H}\}$ and $S = (z_1, \ldots, z_m)$. The **empirical Rademacher complexity** is:

$$\widehat{\mathfrak{R}}_S(\mathcal{F}) \;=\; \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i f(z_i)\right]$$

where $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_m)$ are independent Rademacher variables.

Population Rademacher Complexity

$$\mathfrak{R}_m(\mathcal{F}) \;=\; \mathbb{E}_{S \sim \mathcal{D}^m}\big[\widehat{\mathfrak{R}}_S(\mathcal{F})\big]$$

Intuition: A rich $\mathcal{F}$ can always find f correlated with random labels $\Rightarrow$ high Rademacher complexity.

Generalization Bound via Rademacher Complexity

Theorem 6 ((Rademacher Generalization Bound — Thm. 26.13))

Suppose $\ell(h, z) \in [0, 1]$ *for all* h, z. *Let* $\mathcal{F} = \{\ell(h, \cdot) : h \in \mathcal{H}\}$. *With probability* $\geq 1 - \delta$ *over an i.i.d. sample of size* m*:*

$$\sup_{h \in \mathcal{H}} \left| L_{\mathcal{D}}(h) - L_S(h) \right| \;\leq\; 2\,\mathfrak{R}_m(\mathcal{F}) \;+\; \sqrt{\frac{\ln(2/\delta)}{2m}}$$

Corollary (one-sided)

For the ERM solution $\hat{h}$, with probability $\geq 1 - \delta$:

$$L_{\mathcal{D}}(\hat{h}) \;\leq\; \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + 4\,\mathfrak{R}_m(\mathcal{F}) + 2\sqrt{\frac{\ln(2/\delta)}{2m}}$$

Key message

Rademacher complexity provides *data-dependent*, *distribution-sensitive* generalization bounds. Tighter than VC bounds in many cases.

Properties of Rademacher Complexity

Massart's Lemma

If $\mathcal{F}$ is finite with $|f(z)| \leq B$ for all f, z:

$$\mathfrak{R}_m(\mathcal{F}) \;\leq\; B\sqrt{\frac{2\log|\mathcal{F}|}{m}}$$

- **Composition rules**: $\mathfrak{R}_m(\mathcal{F})$ is preserved under Lipschitz composition (Ledoux-Talagrand contraction).
- **For linear classes**: $\mathcal{H} = \{x \mapsto \langle w, x\rangle : \|w\| \leq B\}$ with $\|x\| \leq R$ a.s.:

$$\mathfrak{R}_m(\mathcal{H}) \;\leq\; \frac{BR}{\sqrt{m}}$$

- **For neural networks**: $\mathfrak{R}_m$ grows with depth and weight norms.

Shattering and VC Dimension

Definition (Shattering — Def. 6.1)

A hypothesis class $\mathcal{H}$ **shatters** a set $C \subseteq \mathcal{X}$ if for every labeling $y : C \to \{0, 1\}$ there exists $h \in \mathcal{H}$ consistent with y:

$$\forall A \subseteq C,\ \exists h \in \mathcal{H} :\ \{x \in C : h(x) = 1\} = A$$

Definition (VC Dimension — Def. 6.5)

The **VC-dimension** of $\mathcal{H}$ is the size of the largest set shattered by $\mathcal{H}$:

$$\mathrm{VCdim}(\mathcal{H}) = \max\big\{|C| : \mathcal{H} \text{ shatters } C\big\}$$

If $\mathcal{H}$ shatters arbitrarily large sets, $\mathrm{VCdim}(\mathcal{H}) = \infty$.

The Fundamental Theorem of Statistical Learning

Theorem 7 ((Fundamental Theorem of Learning — Thm. 6.7))

Let $\mathcal{H}$ be a binary classification class with $\mathrm{VCdim}(\mathcal{H}) = d < \infty$. Then $\mathcal{H}$ is agnostic PAC learnable with sample complexity:

$$\Omega\left(\frac{d + \log(1/\delta)}{\varepsilon^2}\right) \leq m_{\mathcal{H}}(\varepsilon, \delta) \leq O\left(\frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon^2}\right)$$

Moreover, the following are equivalent:

1. *$\mathcal{H}$ is agnostic PAC learnable*
2. *$\mathcal{H}$ has the uniform convergence property*
3. $\mathrm{VCdim}(\mathcal{H}) < \infty$

Consequence: VC dimension completely characterises learnability in binary classification!

Generalization Bound via VC Dimension

Theorem 8 ((VC Generalization Bound))

Let $\mathcal{H}$ have $\mathrm{VCdim}(\mathcal{H}) = d$. With probability $\geq 1 - \delta$ over an i.i.d. sample S of size m, for all $h \in \mathcal{H}$:

$$L_{\mathcal{D}}(h) \;\leq\; L_S(h) \;+\; \sqrt{\frac{8d\ln(em/d) + 8\ln(4/\delta)}{m}}$$

- Bound *decreases* in m: more data $\Rightarrow$ better generalization.
- Bound *increases* in d: richer class $\Rightarrow$ harder to generalize.
- **Structural Risk Minimization**: choose $\mathcal{H}$ to trade off the two terms.

Structural Risk Minimization (SRM)

Decompose $\mathcal{H} = \bigcup_{n \geq 1} \mathcal{H}_n$ with increasing complexity. Choose n that minimizes:

$$L_S(h) + \varepsilon_n(m, \delta)$$

where ε_n is the generalization gap for $\mathcal{H}_n$.

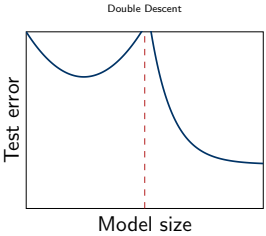
Overfitting within the PAC Framework

When does overfitting occur?

- $\mathrm{VCdim}(\mathcal{H}) \gg m$: more capacity than data.
- ERM finds $\hat{h}$ with $L_S(\hat{h}) = 0$ but $L_{\mathcal{D}}(\hat{h}) \approx 1/2$.
- Generalization gap $= L_{\mathcal{D}}(\hat{h}) - L_S(\hat{h})$ is large.

Double descent (modern view)

In overparameterized models ($p \gg n$), test error can *decrease again* after interpolation threshold (neural nets, random forests). Classical U-shaped curve is not the whole story.



Sample Splitting: Train / Validation / Test

Why split?

We cannot evaluate generalization on the same data used for training:

- Training error is a *biased* estimate of true risk.
- Hyperparameter selection on test set leads to *adaptive overfitting*.

Hold-out / Train–Test split

Split S into S_{train} ($1-\alpha$ fraction) and S_{test} (α fraction):

$$\hat{h} = \mathrm{ERM}_{\mathcal{H}}(S_{\text{train}}), \qquad \widehat{L_{\mathcal{D}}}(\hat{h}) = L_{S_{\text{test}}}(\hat{h})$$

$L_{S_{\text{test}}}(\hat{h})$ is an unbiased estimate of $L_{\mathcal{D}}(\hat{h})$.

Three-way split

Train: fit model parameters. **Validation**: select model / hyperparameters. **Test**: report final generalization error (use *once*).

k-Fold Cross-Validation

Procedure

1. Partition S into k equal folds $S_1, \ldots, S_k$.
2. For $j = 1, \ldots, k$: train on $S \setminus S_j$, evaluate on S_j.
3. Report $\widehat{\mathrm{CV}}_k = \frac{1}{k} \sum_{j=1}^{k} L_{S_j}(\hat{h}^{(-j)})$.

Advantages:

- Uses all data for both training and evaluation.
- Reduces variance of generalization estimate.
- $k = n$ (leave-one-out): nearly unbiased.

Disadvantages:

- $k\times$ training cost.
- CV estimate can still overfit if used for model selection many times.

No Free Lunch Theorem

Theorem 9 ((No Free Lunch — Thm. 5.1))

Let $\mathcal{A}$ be any learning algorithm for binary classification over $\mathcal{X}$ (with $|\mathcal{X}| = \infty$). For any $m < |\mathcal{X}|/2$ and any hypothesis $h : \mathcal{X} \to \{0, 1\}$, there exists a distribution $\mathcal{D}$ such that:

$$\mathbb{P}_{S \sim \mathcal{D}^m}\left[L_{\mathcal{D}}(\mathcal{A}(S)) \geq 1/8\right] \;\geq\; 1/7$$

Implication

No learner succeeds uniformly over all distributions. **Inductive bias is unavoidable**: we must restrict $\mathcal{H}$ or impose structure.

In practice: choose $\mathcal{H}$ informed by domain knowledge, prior work, and **validated on held-out data**.

Summary: Key Concepts

Statistical framework:

- Risk $L_{\mathcal{D}}(h) = \mathbb{E}_{\mathcal{D}}[\ell(h, z)]$
- Empirical risk $L_S(h) = \frac{1}{m}\sum_i \ell(h, z_i)$
- ERM: $\hat{h} = \arg\min_{h \in \mathcal{H}} L_S(h)$
- PAC learning, sample complexity

Overfitting & complexity:

- Bias–variance tradeoff
- Uniform convergence
- SRM

Complexity measures:

- VC dimension: learnability $\Leftrightarrow \mathrm{VCdim} < \infty$
- Sauer–Shelah: polynomial growth
- Rademacher complexity: data-dependent bounds

Avoiding overfitting:

- Regularization / restrict $\mathcal{H}$
- Train/validation/test split
- k-fold cross-validation

Next lecture

Linear models, regularization (Ridge, LASSO), and their statistical guarantees.

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